

Monte Carlo Methods

with Applications in Finance

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Introduction

- A derivative is a security whose value depends on an underlying asset, a group of assets, or a benchmark.
- Analytical solutions exist for simple derivatives (e.g., Black-Scholes formula for European call options).
- Complex derivatives (path-dependent or multi-asset) are often impossible to solve analytically.

- Monte Carlo methods estimate expected payoffs by simulating many random scenarios.
- Well-suited for path-dependent and multi-asset derivatives.
- Advantages: robust, flexible, handles high-dimensional problems.
- Limitations: slow convergence, high computational cost for precise estimates.

Option Pricing

Suppose the payoff of a derivative is a function $h(X)$ of a random variable X related to the underlying asset.

By the principle of no-arbitrage[4, 1], the derivative's price is the expected discounted payoff under a risk-neutral measure:

$$v = \mathbb{E}[e^{-rT}h(X)],$$

where

- r is the *risk-free interest rate*,
- T is the time to maturity.

Under this measure, all assets are expected to grow at the risk-free rate, reducing pricing to computing expected discounted payoffs.

The underlying asset price is commonly modeled as a geometric Brownian motion:

$$S_t = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right), \quad W_t \sim N(0, t),$$

where

- S_0 is the initial price,
- σ is the volatility (constant), and
- W_t is a standard Brownian motion.

This model will be assumed throughout this report.

Examples of Derivative Payoffs

1. **European Call Option:** Strike K , underlying S_T , discounted payoff:

$$h(S_T) = e^{-rT}(S_T - K)^+, \quad v = \mathbb{E}[h(S_T)].$$

Closed-form solution via Black-Scholes formula[1]:

$$v = S_0 \Phi(\sigma\sqrt{T} - \theta) - Ke^{-rT} \Phi(-\theta),$$

with

$$\theta = \frac{1}{\sigma\sqrt{T}} \ln \frac{K}{S_0} + \left(\frac{\sigma}{2} - \frac{r}{\sigma}\right)\sqrt{T}.$$

2. **Asian Call Option:** Discretely monitored average price option, payoff:

$$h(S_{t_1}, \dots, S_{t_m}) = e^{-rT} \left(\frac{1}{m} \sum_{i=1}^m S_{t_i} - K \right)^+.$$

Closed-form solutions exist only in special cases (e.g. geometric average price option); and no analytical solution is available [3].

Monte Carlo methods can be used to estimate v by approximating the expected payoff through random sampling.

Plain Monte Carlo Method

Problem: Estimate the quantity $v = \mathbb{E}[X]$, where $X = h(Z)$ is a function of a r.v. Z .

Plain Monte Carlo estimator:

- Generate N i.i.d. samples $Z_i \sim f_Z, i = 1, \dots, N$.
- Compute the sample mean

$$\hat{v} = \frac{1}{N} \sum_{i=1}^N h(Z_i).$$

- $\hat{v}$ is unbiased:

$$\mathbb{E}[\hat{v}] = v, \quad \text{Var}(\hat{v}) = \frac{\text{Var}(X)}{N}.$$

As N increases, $\hat{v}$ converges to the true mean v .

Examples: European Call Option

European Call Option: Strike $K = 60$, maturity $T = 1$ year. Discounted payoff:

$$h(S_T) = e^{-rT}(S_T - K)^+.$$

Parameters: $S_0 = 50$, $r = 0.05$, $\sigma = 0.2$ drive the Geometric Brownian Motion.

N	10^2	10^3	10^4	10^5	10^6
Estimate	1.8824	1.6428	1.6586	1.6119	1.6181
S.E.	0.6072	0.1388	0.0447	0.0135	0.0043
R.E.	32.25%	8.45%	2.69%	0.84%	0.27%

Table 1: European Call Option: Monte Carlo estimates for different sample sizes N

Note. Convergence to $v = 1.6237$ as N increases.

Examples: Asian Call Option

Asian Call Option: Discretely monitored with $m = 100$ time points, strike $K = 60$, maturity $T = 1$, discounted payoff:

$$h(S_{t_1}, \dots, S_{t_m}) = e^{-rT} \left(\frac{1}{m} \sum_{i=1}^m S_{t_i} - K \right)^+.$$

Parameters: $S_0 = 50, r = 0.05, \sigma = 0.2$.

N	10^2	10^3	10^4	10^5	10^6
Estimate	0.2823	0.2534	0.2650	0.2715	0.2700
S.E.	0.1224	0.0357	0.0122	0.0040	0.0013
R.E.	43.36%	14.09%	4.60%	1.47%	0.47%

Table 2: Asian Call Option: Monte Carlo estimates for various N

Rate of Convergence

By the *Central Limit Theorem*, for large N :

$$\sqrt{N}(\hat{v} - v) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

where $\sigma^2 = \text{Var}(X)$. So convergence is **slow**. $\text{S.E.}(\hat{v}) = \mathcal{O}(N^{-1/2})$.

- High variance σ^2 increases with problem dimension or payoff complexity¹.
- Large sample sizes needed for high precision, computationally expensive.

¹Example: Deep out-of-the-money options lead to skewed payoff distributions and large relative errors.

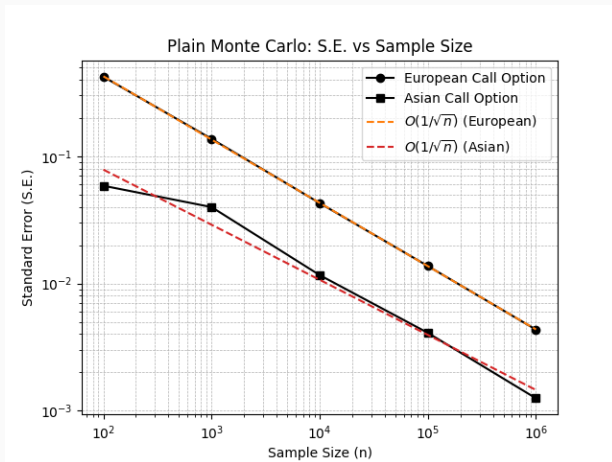


Figure 1: Plain Monte Carlo: Standard Error (S.E.) vs Sample Size N

Motivation for Variance Reduction

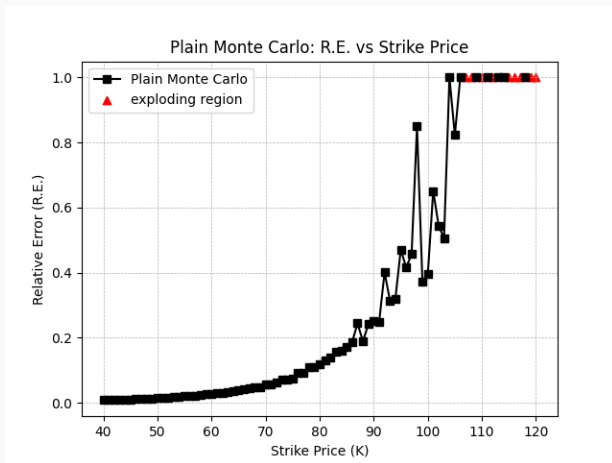


Figure 2: Plain Monte Carlo: R.E. versus Strike Price, $N = 10,000$

- Deep out-of-the-money option when $K \gg S_0$. Frequency of exercising low. Variance σ^2 high. Slow convergence.

- Increasing N to reduce standard error is expensive: a 10x reduction in S.E. requires 100x more samples.
- **Variance reduction techniques** aim to lower the estimator variance σ^2 without increasing N .
- Popular methods: Antithetic sampling, Control variates, Stratified sampling, Importance sampling, **Control functionals** (recent approach)
- Other advanced methods: quasi-Monte Carlo², multi-level Monte Carlo.

²Quasi-Monte Carlo replaces random samples with deterministic low-discrepancy sequences (e.g., Sobol', Halton), reducing integration error without relying on randomness.

Variance Reduction Techniques

Control Variates: Concept

Suppose a r.v. Y exists such that:

- i. X and Y are correlated;
- ii. generating Y_i from X_i is cheap;
- iii. the expected value $\mathbb{E}[Y]$ is known exactly.

Control Variate (CV) Estimator:

$$\hat{v}_{CV} = \bar{X} - b(\bar{Y} - \mathbb{E}[Y]),$$

where $\bar{X}$, $\bar{Y}$ are sample means, and b is chosen to minimize variance³:

$$b^* = \frac{\text{Cov}[X, Y]}{\text{Var}[Y]}.$$

Variance of the estimator:

$$\text{Var}[\hat{v}_{CV}] = (1 - \beta^2) \frac{\text{Var}[X]}{N},$$

where β is the correlation between X and Y .

³Typically, b^* is computed from a smaller pilot sample.

Asian Option: Control Variates vs Plain MC

Asian Call Option: Discretely monitored with $m = 100$ time points

$0 < t_1 < \dots < t_m = T$, maturity $T = 1$, discounted payoff:

$$X = h(S_{t_1}, \dots, S_{t_m}) = e^{-rT} (\bar{S} - K)^+, \quad \bar{S} = \frac{1}{m} \sum_{i=1}^m S_{t_i}.$$

Parameters: $S_0 = 50$, $r = 0.05$, $\sigma = 0.2$, $T = 1$, and $N = 10,000$.

Control Variate: Average price call option with geometric mean. Discounted payoff:

$$Y = e^{-rT} (\bar{S}_G - K)^+, \quad \bar{S}_G = \left(\prod_{i=1}^m S_{t_i} \right)^{1/m}.$$

The mean is known, see *Example 2.5.* in [7]:

$$\bar{\mu} = \log S_0 + \left(r - \frac{\sigma^2}{2}\right) \frac{1}{m} \sum_{i=1}^m t_i,$$

$$\bar{\sigma}^2 = \frac{\sigma^2}{m^2} \sum_{i=1}^m (2m - 2i + 1) t_i.$$

$$E[Y] = e^{-rT} [e^{\bar{\mu} + \frac{1}{2} \bar{\sigma}^2} \Phi(\bar{\sigma} - \theta) + K \Phi(-\theta)], \quad \theta = (\log K - \bar{\mu}) / \bar{\sigma}$$

Strike Price	K = 60		K = 70		K = 80	
	Plain MC	CV	Plain MC	CV	Plain MC	CV
Estimate	0.2887	0.2720	0.0159	0.0107	0.0000	0.0009
S.E.	0.0130	0.0009	0.0032	0.0005	0.0000	0.0001
R.E.	4.51%	0.33%	19.83%	4.26%	nan%	13.39%
Variance Reduction	99.53%		97.92%		–	

Table 3: Asian Call Option: Plain Monte Carlo (Plain MC) vs Control Variates (CV). Reported values include estimates, standard errors (S.E.), relative errors (R.E.), and variance reduction.

Straddle Option: Control Variates (Linear)

Straddle Option: Strike price K , underlying stock S_t , discounted payoff

$$X = h(S_T) = e^{-rT}[(S_T - K)^+ + (K - S_T)^+]$$

Parameters: $S_0 = 50$, $r = 0.02$, $\sigma = 0.2$, $T = 1$, $K = 50$ and $N = 10,000$

Control Variate: Stock price at maturity.

$$Y = S_T \Rightarrow \mathbb{E}[Y] = S_0 e^{rT}$$

Initial Price	$S_0 = 40$		$S_0 = 50$		$S_0 = 60$	
	Plain MC	CV	Plain MC	CV	Plain MC	CV
Estimate	10.3312	10.4381	7.8795	7.8737	12.7905	12.7598
S.E.	0.0614	0.0400	0.0628	0.0593	0.1018	0.0416
R.E.	0.59%	0.38%	0.80%	0.75%	0.83%	0.33%
$\hat{\beta}^2$	57.54%		10.89%		83.34%	

Table 4: Straddle Option using linear control variate $Y = S_T$.

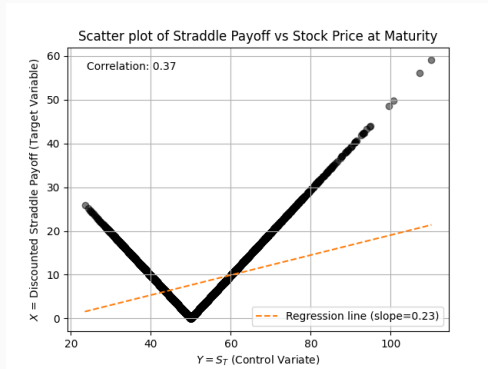


Figure 3: $h(S_T)$ vs S_T for Straddle Option, $S_0 = K = 50$, using $Y = S_T$ as a control variate.

- Highly non-linear relationship between X and Y when $S_0 = K$.
- Reduced effectiveness of linear CV.

Straddle Option: Control Variates (Quadratic)

Use two control variates: $Y_1 = S_T$, $Y_2 = S_T^2$. Then $\mathbb{E}[S_T^2] = E[S_T]^2 + \text{Var}(S_T) = S_0^2 e^{(2r+\sigma^2)T}$ so $E[Y_1]$ and $E[Y_2]$ are known.

New Control Variate Estimator:

$$\hat{v}_{CV} = \bar{X} - b_1^*(\bar{Y}_1 - \mathbb{E}[Y_1]) - b_2^*(\bar{Y}_2 - \mathbb{E}[Y_2]).$$

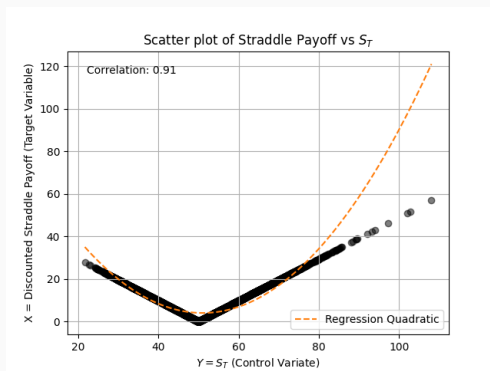


Figure 4: $h(S_T)$ vs S_T with $S_0 = K = 50$, using S_T and S_T^2 as control variates.

Remark 1. b_1^* and b_2^* computed via linear regression $X \sim Y_1 + Y_2$.

Remark 2. Adding quadratic term improves variance reduction significantly.

	Plain	CV	Better CV
Estimate	7.8795	7.8737	7.9489
S.E.	0.0638	0.0593	0.0252
R.E.	0.80%	0.75%	0.32%
$\hat{\beta}^2$		10.89%	83.93%

Table 5: Straddle Option: comparison of linear and quadratic CV.

Generalization:

- Control Variate as linear combinations of features

$$Y' = \sum_{i=1}^m b_i \psi_i(Y)$$

b_i chosen to minimize variance.

- Choice of functions ψ_i in $\Phi(Y) = (\psi_1(Y), \dots, \psi_m(Y))$ affects efficiency of variance reduction.
- Require $\mathbb{E}[Y]$ to be known exactly.
- β may be small $\rightarrow$ CV is less effective.

Key Idea

Replace linear combinations of basis functions with a *rich function space* for variance reduction.

Control Functionals: Overview

Setup: estimate $v = \mathbb{E}[h(Z)]$, where $Z \sim p(z)$.

Control Variates: approximates $h(z)$ by

$$h(z) \approx g(z) = b_1\psi_1(z) + \cdots + b_m\psi_m(z),$$

- $\mathbb{E}[\psi_i(Z)]$ are known.
- Coefficients b_i minimize the empirical error between $h(z)$ and $g(z)$ on Z_i .

Control Functionals[5]: approximates $h(z)$ by

$$h(z) \approx g(z) \in \mathcal{H}_+ \gg \text{span}(\psi_1, \dots, \psi_m)$$

- $\mathcal{H}_+$ is a *Reproducing Kernel Hilbert Space* (RKHS) with a symm., p.d., kernel $k_+(\cdot, \cdot)$.
- Regularized Empirical Risk Functional:

$$g^* = \arg \min_{g \in \mathcal{H}_+} \frac{1}{N} \sum_{i=1}^N \|h(Z_i) - g(Z_i)\|^2 + \lambda \|g\|_{\mathcal{H}_+}^2,$$

where $\lambda > 0$ is a regularization parameter controlling smoothness of g .

Representer Theorem:

Any minimizer g^* of the *regularized empirical risk functional* admits a representation of the form

$$g^*(z) = \sum_{i=1}^N b_i k_+(z, Z_i),$$

for some coefficients $b_i \in \mathbb{R}$, [6].

CF Estimator:

$$\hat{v}_{\text{CF}} = \frac{1}{N} \sum_{i=1}^N \underbrace{h(Z_i) - g^*(Z_i)}_{\text{residuals}} + \mathbb{E}[g^*(Z)].$$

- Construct k_+ such that $\mathbb{E}[k_+(Z, Z_i)]$ is known or computed easily.

Stein Operator[2]

Given density p and function ϕ , the *Stein Operator* $\mathcal{S}_p$ is defined as

$$\mathcal{S}_p[\phi](z) = \nabla_z \cdot \phi(z) + \phi(z) \cdot \nabla_z \log p(z).$$

- **Zero-Mean Property** If X has density p , then $\mathbb{E}[\mathcal{S}_p[\phi](X)] = 0$.
- Apply $\mathcal{S}_p$ to both arguments of $k(\cdot, \cdot)$ to obtain a “zero-mean kernel” $k_0(\cdot, \cdot)$.

Modified RBF Kernel: The standard normal density p and a Radial Basis Function Kernel *modified with a decay term*

$$k(z, z') = (1 + \alpha_1 \|z\|^2 + \alpha_1 \|z'\|^2)^{-1} \exp\left(-\frac{\|z - z'\|^2}{2\alpha_2^2}\right),$$

where $\alpha_1, \alpha_2 > 0$ are the decay scale and length scale terms respectively. Then⁴

$$k_0(z, z') = k(z, z') \left(\frac{1 + \|z - z'\|^2}{\alpha_2^2} + \frac{\|z - z'\|^2}{\alpha_2^4} + \frac{2\alpha_1(\|z - z'\|^2 + 2zz')}{1 + \alpha_1\|z\|^2 + \alpha_1\|z'\|^2} + 2zz' \right).$$

⁴See appendix for derivation.

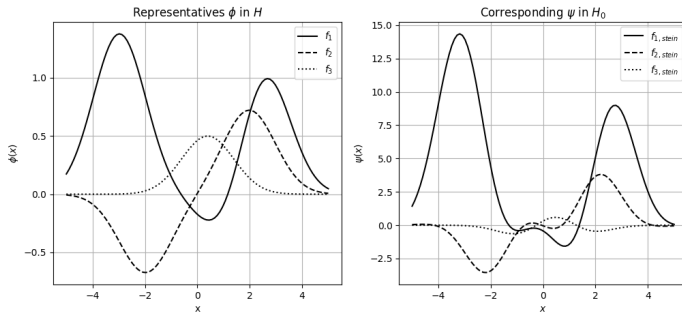


Figure 5: Elements of $\mathcal{H}$ and $\mathcal{H}_0$ for the standard normal distribution and modified RBF kernel with decay scale $\alpha_1 = 0.001$ and length scale $\alpha_2 = 1.0$

Step 1: (Sample Splitting) Draw N i.i.d. samples $\{Z_i\}_{i=1}^N$ from $p(z)$ and split into:

- *Training set:* $\mathcal{D}_0 = \{Z_i\}_{i=1}^M \rightarrow$ “pilot sample”
- *Test set:* $\mathcal{D}_1 = \{Z_i\}_{i=M+1}^N$

Step 2: (Learn Control Functional) Solve the regularized least squares (RLS) problem for $g^* \in \mathcal{H}_+$ on $\mathcal{D}_0$:

$$g^* = \arg \min_{g \in \mathcal{H}_+} \frac{1}{M} \sum_{i=1}^M \|h(Z_i) - g(Z_i)\|^2 + \lambda \|g\|_{\mathcal{H}_+}^2 \quad (1)$$

Parameters: Choose M such that $M/N \rightarrow \eta \in (0, 1)$ and $\lambda = O(M^{-1/2})$.

Step 3: (Construct Estimator) The control functional estimator on $\mathcal{D}_1$ is computed as:

$$\hat{v}_{\text{CF}} = \frac{1}{N-M} \mathbf{1}^T (\mathbf{h}_1 - \hat{\mathbf{h}}_1) + \underbrace{\frac{\mathbf{1}^T (\mathbf{K}_0 + \lambda \text{MI})^{-1} \mathbf{h}_0}{1 + \mathbf{1}^T (\mathbf{K}_0 + \lambda \text{MI})^{-1} \mathbf{1}}}_{\mathbb{E}[g^*(Z)]}$$

where

$$\begin{aligned} \mathbf{h}_0 &= [h(Z_1), \dots, h(Z_M)]^T, \quad \mathbf{h}_1 = [h(Z_{M+1}), \dots, h(Z_N)]^T, \\ \mathbf{1} &= [1, \dots, 1]^T, \quad (\mathbf{K}_0)_{i,j} = k_0(Z_i, Z_j). \end{aligned}$$

The predictions on $\mathcal{D}_1$ are

$$\hat{\mathbf{h}}_1 = \mathbf{K}_{1,0} (\mathbf{K}_0 + \lambda \text{MI})^{-1} \mathbf{h}_0 + (1 - \mathbf{K}_{1,0} (\mathbf{K}_0 + \lambda \text{MI})^{-1} \mathbf{1}) \frac{\mathbf{1}^T (\mathbf{K}_0 + \lambda \text{MI})^{-1} \mathbf{h}_0}{1 + \mathbf{1}^T (\mathbf{K}_0 + \lambda \text{MI})^{-1} \mathbf{1}},$$

with $(\mathbf{K}_{1,0})_{i,j} = k_0(Z_{M+i}, Z_j)$.

Super-root-N convergence.

Under regularity conditions on the kernel $k_0(\cdot, \cdot)$ and the density p , $\hat{v}_{\text{CF}}$ achieves *super-root-N convergence*[5].

Straddle Option: Control Functionals

Straddle Option: Revisit the straddle option with $N = 10,000$.

CF Estimator: *modified RBF kernel* with parameters ($\alpha_1 = 0.001, \alpha_2 = 1, \eta = 0.25$).

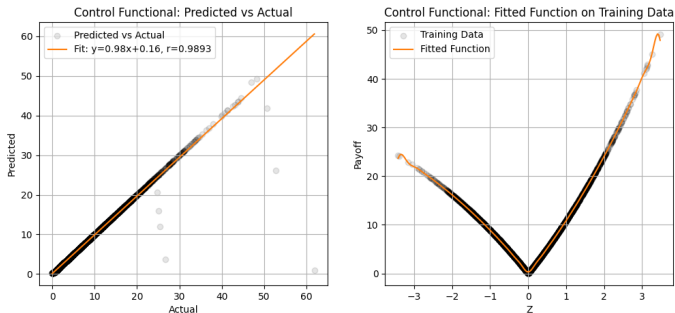


Figure 6: Straddle Option Pricing: Control Functionals. High correlation between h_1 and $\hat{h}_1$ (left). Learned g^* fits h well (right).

Remark 3. CF Estimator outperforms the quadratic CV estimator.

	Plain	CV	Better CV	CF	Theoretical
Estimate	7.8795	7.8737	7.9489	7.9251	7.9260
S.E.	0.0638	0.0593	0.0252	0.0055	-
R.E.	0.80%	0.75%	0.32%	0.07%	-
$\hat{\beta}^2$	-	10.89%	83.93%	-	-

Table 6: Straddle Option: comparison of linear and quadratic CV with CF estimator.

Butterfly Multistrike Option: Control Functionals

Butterfly Multistrike Option: Deep out-of-the-money option. Discounted payoff:

$$h(S_T) = e^{-rT}[(S_T - K_1)^+ - 2(S_T - K_2) + (S_T - K_3)^+].$$

Parameters: Strike prices: $K_1 = 90, K_2 = 100, K_3 = 120$

$S_0 = 50, r = 0.05, \sigma = 0.3, T = 1$, and $N = 10,000$.

CF estimator: *modified RBF kernel* with parameters ($\alpha_1 = 0.001, \alpha_2 = 0.25, \eta = 0.4$)

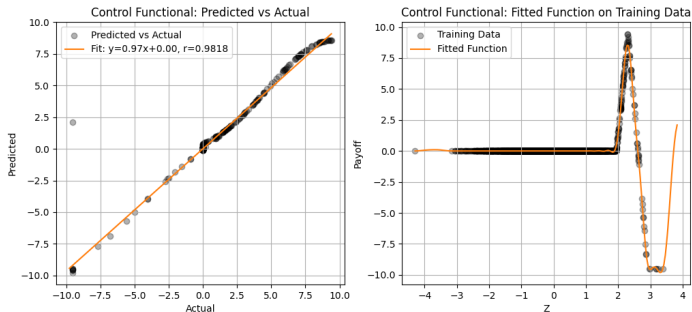


Figure 7: Butterfly Multistrike Option Pricing: Control Functionals. High correlation between h_1 and $\hat{h}_1$ (left). Learned g^* fits h well (right).

CV Estimator: Same as before. $Y = S_T$

	Plain	CV	CF	Theoretical
Estimate	0.0697	0.0687	0.0685	0.0683
S.E.	0.0087	0.0089	0.0020	-
R.E.	12.46%	12.80%	2.93 %	-

Table 7: Butterfly Multistrike Option: Control Functional vs Plain Monte Carlo results

Control Functionals: Rate of Convergence

CF estimator: *modified RBF Kernel* with parameters: ($\alpha_1 = 0.001, \alpha_2 = 0.25, \eta = 0.1$)

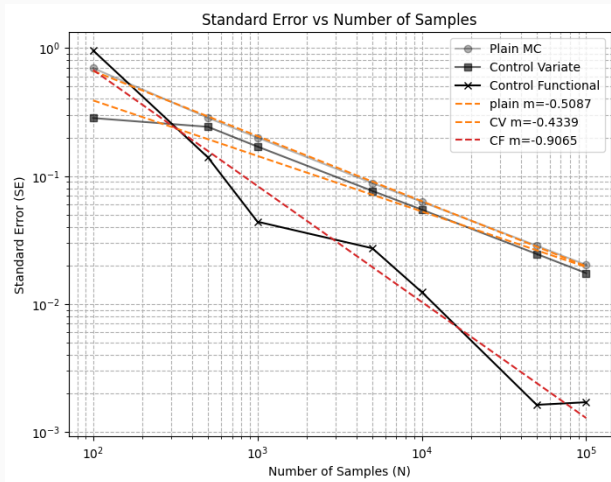


Figure 8: Straddle Option: S.E. versus Sample Size N .

Computational Cost. Solving the RLS problem in (1) requires inverting an $M \times M$ kernel matrix K , with computational cost $\mathcal{O}(M^3) \sim \mathcal{O}(N^3)$ and memory cost $\mathcal{O}(M^2)$. For large M , the variance reduction must be weighed against this cost.

- Case: generating samples $h(X_i)$ expensive.
- Case: generating samples $h(X_i)$ cheap.

Kernel Selection. Performance is sensitive to the choice of k_0 and the regularization parameter λ .

Theoretical Assumptions. Super-root- N convergence is guaranteed under certain conditions on the kernel k_0 , the density p , and the dataset.

- Kernel $k_0(\cdot, \cdot)$ is bounded:

$$\sup_{z \in \Omega} k_0(z, z) < \infty.$$

Does not hold for standard RBF kernel:

$$k(z, z') = \exp\left(-\frac{\|z - z'\|^2}{2\alpha^2}\right).$$

- Density is “nice”; $p \in C^1(\Omega, \mathbb{R})$, not true for many *useful* p , e.g. uniform dist.

Conclusion

Report

Get the full report and code from

`github.com/chinzhening/capstone-project`

An implementation of the Control Functionals Method for Modified RBF Kernel can be found at

`./src/control_functionals_implementation.py`

```
class ControlFunctional:
    """A Control Functional model for variance reduction in
    Monte Carlo integration."""
    def __init__(self, X, Y, key, kernel_fn=
        modified_stein_rbf_kernel, eta=0.10, length_scale=1.0,
        decay_scale=1e-3):
        ...

    def fit(self, alpha=1e-3): ...
    def predict(self): ...
    def summary(self): ...
```

Questions?

Reproducing Kernel Hilbert Space

Definition. A Reproducing Kernel Hilbert Space (RKHS) is a Hilbert space of functions $\mathcal{H}$ in which evaluation at any point is a continuous linear functional.

Reproducing Property. There exists a positive definite kernel function $k(\cdot, \cdot)$ such that for all $f \in \mathcal{H}$ and $z \in \Omega$, we have

$$f(z) = \langle f, k(\cdot, z) \rangle_{\mathcal{H}}.$$

Construction of $k_+(\cdot, \cdot)$ via Stein Operators

Step 1. $k(\cdot, \cdot)$ to $k_0(\cdot, \cdot)$:

$$k_0(z, z') = \mathcal{S}_p^z \mathcal{S}_p^{z'} [k(z, z')]$$

This can be expressed explicitly as

$$k_0(z, z') = \nabla_z \cdot \nabla_{z'} k(z, z') + u(z) \cdot \nabla_{z'} k(z, z') + u(z') \cdot \nabla_z k(z, z') + u(z) \cdot u(z') k(z, z').$$

where $u(x) = \nabla_x \cdot \log p(x)$ is the score function of p . **Note.** Kernel k_0 is p.d. [5] and satisfies zero-mean property:

$$\int_{\Omega} k_0(z, z') \cdot p(z') dz' = 0.$$

Step 2. $k_0(\cdot, \cdot)$ to $k_+(\cdot, \cdot)$:

Add space of constant functions $\mathcal{C} = \{c : c \in \mathbb{R}\}$ with associated kernel $k_{\mathcal{C}}(z, z') = 1$

$$\mathcal{H}_+ = \mathcal{H}_0 + \mathcal{H}_{\mathcal{C}}$$

$$k_+(z, z') = 1 + k_0(z, z').$$

Then $\mathcal{H}_+$ is also a RKHS. Solving the RLS problem in (1) is equivalent to **Regularized Kernel Regression** with $k_+(\cdot, \cdot)$.

Modified RBF Kernel $k(\cdot, \cdot) \rightarrow k_0(\cdot, \cdot)$

Base modified RBF kernel:

$$k(z, z') = \frac{\exp(-\|z - z'\|^2 / 2\alpha_2^2)}{1 + \alpha_1(\|z\|^2 + \|z'\|^2)}, \quad \alpha_1, \alpha_2 > 0$$

Standard Normal Density

$$p(z) = (2\pi)^{-1} \exp(-z^2/2)$$

Step 1. Score function for standard normal:

$$u(z) = \nabla_z \log p(z) = -z \Rightarrow \mathcal{S}_p^z[\phi(z)] = \nabla_z \cdot \phi(z) - z \cdot \phi(z)$$

Step 2. Apply Stein Operator to left argument z .

$$k_0(z, z') = \mathcal{S}_p^{z'} \mathcal{S}_p^z[k(z, z')]$$

$$\begin{aligned} \phi(z) = k(z, \cdot) \Rightarrow \nabla_z \cdot \phi(z) &= -\frac{\exp(-\|z - z'\|^2 / 2\alpha_2^2)}{1 + \alpha_1\|z\|^2 + \alpha_2\|z'\|^2} \left(\frac{z - z'}{\alpha_2^2} - \frac{2\alpha_1 z}{1 + \alpha_1\|z\|^2 + \alpha_1\|z'\|^2} \right) \\ &= -k(z, z') \left(\frac{z - z'}{\alpha_2^2} - \frac{2\alpha_1 z}{1 + \alpha_1\|z\|^2 + \alpha_1\|z'\|^2} \right) \end{aligned}$$

Applying on the left.

$$\begin{aligned}\mathcal{S}_\rho^z[k(z, z')] &= -k(z, z') \left(\frac{z - z'}{\alpha_2^2} - \frac{2\alpha_1 z}{1 + \alpha_1 \|z\|^2 + \alpha_1 \|z'\|^2} \right) - z \cdot k(z, z') \\ &= -k(z, z') \left(\frac{z - z'}{\alpha_2^2} + \frac{2\alpha_1 z}{1 + \alpha_1 \|z\|^2 + \alpha_1 \|z'\|^2} + z \right)\end{aligned}$$

Step 3. Apply Stein Operator to right argument z' .

$$\begin{aligned}\nabla_{z'} \cdot \mathcal{S}_\rho^z[k(z, z')] &= -k(z, z') \left(-\frac{1}{\alpha_2^2} - \frac{4\alpha_1^2 z z'}{(1 + \alpha_1 \|z\|^2 + \alpha_1 \|z'\|^2)^2} \right) \\ &\quad + k(z, z') \left(\frac{z' - z}{\alpha_2^2} - \frac{2\alpha_1 z'}{1 + \alpha_1 \|z\|^2 + \alpha_1 \|z'\|^2} \right) \\ &= k(z, z') \left(\frac{1 + z' - z}{\alpha_2^2} + \frac{4\alpha_1 z z' - 2\alpha_1 z' (1 + \alpha_1 \|z\|^2 + \alpha_1 \|z'\|^2)}{(1 + \alpha_1 \|z\|^2 + \alpha_1 \|z'\|^2)^2} \right)\end{aligned}$$

Apply to right argument z' :

$$\begin{aligned}\mathcal{S}_p^{z'} \mathcal{S}_p^z[k(z, z')] &= \mathcal{S}_p^{z'} \left[-k(z, z') \left(\frac{z - z'}{\alpha_2^2} + \frac{2\alpha_1 z}{1 + \alpha_1 \|z\|^2 + \alpha_1 \|z'\|^2} + z \right) \right] \\ &= \dots \\ &= k(z, z') \left(\frac{1 + \|z - z'\|^2}{\alpha_2^2} + \frac{\|z - z'\|^2}{\alpha_2^4} + \frac{2\alpha_1 (\|z - z' \|^2 + 2zz')}{1 + \alpha_1 \|z\|^2 + \alpha_1 \|z'\|^2} + zz' \right)\end{aligned}$$

Control Functional Cost

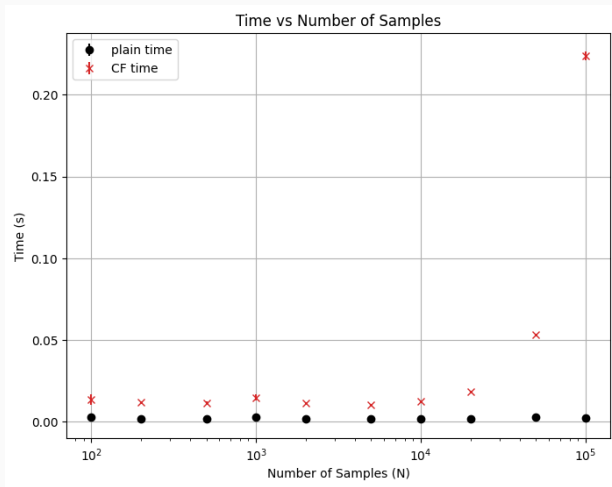


Figure 9: Straddle Option: Computational Cost (time) versus Sample Size N



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